

MD KHALEQUZZAMAN CHOWDHURY SAYEM

Current Position: Researcher at UNIST, South Korea

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Research Interests

Vision-Language Models, Multimodal Reasoning, 3D Human-Object Interaction, Continual Learning, Generative Models.

EDUCATION

Ulsan National Institute of Science & Technology

Sep 2023 – August 2025

Masters in Computer Science and Engineering

Ulsan, Korea

Ulsan National Institute of Science & Technology

Sep 2019 – Aug 2023

Bachelor in Industrial Engineering

Ulsan, Korea

RELEVANT PUBLICATIONS AND PROJECT (* denotes equal contribution)

- E. Ismayilzada*, **K. Sayem***, Y. Y. Tiruneh, M. T. Chowdhury, M. Boboev, and S. Baek. QORT-Former: Query-optimized Real-time Transformer for Understanding Two Hands Manipulating Objects. **Accepted at AAAI 2025**. Project Page: <https://kcsayem.github.io/QORT-Former/>
 - Co-first author on a paper detailing the development of a novel Transformer-based framework (QORT-Former) that achieves real-time 3D pose estimation for two hands and an object with a real-time speed of 53.5 FPS.
 - Achieved state-of-the-art accuracy, outperforming previous methods by significant margins on the H2O dataset (up to +27.2%) and FPHA dataset (up to +10.4%).
 - Proposed a three-step feature update process and integrated contact map features to improve accuracy while using a minimal number of queries and a single decoder.
- **HandVQA**: Diagnosing Fine-Grained Spatial Reasoning Failures in Vision-Language Models via Hand Pose Question Answering. (**Masters Thesis/Under review at NeurIPS**) [**Download PDF**]
 - Authored a Master's Thesis presenting **HandVQA**, a large-scale diagnostic benchmark of over 1.6 million questions to evaluate vision-language models' spatial reasoning on fine-grained hand poses.
 - Designed and implemented an automated pipeline to generate questions probing relationships between hand joints, such as angles, distances, and relative positions.
 - Demonstrated that while fine-tuning significantly improves model performance, base models exhibit systematic biases and perform at or below random guessing on spatial reasoning tasks.
- **K. Sayem**, E. Fozilov, M. Chowdhury, "4D Instruct-GS2G: Editing 4D Gaussians using text instructions". The project was conducted during the 3D Computer Vision Class at UNIST. (Report, Code)
 - Implemented the integration of text-derived editing for the project, which developed a method to extend semantic editing to dynamic 3D scenes using text instructions.
 - Achieved realistic, globally consistent edits on real-world scenes with fast training times (30 mins per scene) and high rendering speeds (up to 82 FPS).

RELEVANT WORK EXPERIENCES

Vision & Learning Lab 

Jan 2023 – Present

Research Assistant

UNIST, Ulsan, Korea

- Led the development of **HandVQA**, a diagnostic benchmark of over 1.6 million questions, as part of my Master's Thesis.
- Co-first author and contributor to the **QORT-Former** project, a real-time Transformer for 3D pose estimation, accepted at AAAI 2025.
- Improved AP by 3.7% for Image Segmentation using RGB and Thermal Image Fusion with GAN.

Machine Learning & Vision Lab 

Sep 2021 – Feb 2022

Research Assistant

UNIST, Ulsan, Korea

- Improved accuracy of the **HRNet** 2D Human Pose Estimation model on the **COCO** dataset by 0.1% through augmentation techniques.

Statistical Decision Making Lab

Dec 2020 – Aug 2021

Research Assistant

UNIST, Ulsan, Korea

- Reduced computational cost of a news article recommendation system by parallelizing **bandit algorithms**.
- Achieved significant acceleration (up to 3.48x) in parallelized versions of LinUCB and Thompson Sampling.

TECHNICAL SKILLS

Language/Frameworks: Python, Numpy, Scikit-learn, Pandas, OpenCV, Tensorflow, Pytorch, Networkx, Linux, Git

TEACHING ASSISTANT EXPERIENCES

Introduction to AI Programming in Python

- **Instructor:** Professor Michael Jon Burrell
- **Semesters:** Fall 24, Spring 24, Fall 23

Discrete Mathematics

- **Instructor:** Professor Michael Jon Burrell
- **Semester:** Fall 24

Advanced Computer Vision

- **Instructor:** Professor Seungryul Baek
- **Semester:** Spring 24

LEADERSHIP

PAIS- Partnership for AI and Society

Jan 2021 – Present

President

UNIST

- Initiated a club with the help of **Prof. Bradley Tatar** & students associated with AI research or interested in AI to discuss the risk of AI in our society and how can it be prevented.
- Organized forum and talks to improve awareness on *Data Privacy* among UNIST students.

HONORS & AWARDS

Graduate School Scholarship

Funded By: Korean Government

- The scholarship covers the tuition fees, meal plan & monthly stipends for two years of masters.

UNIST Dream Scholarship

Issued By: UNIST

- The scholarship covers the tuition fees, meal plan & monthly stipends for four years of bachelor.

Acquired \$5K funding through AI Challengers Program(AICP)

Issued By: UNIST Innovative Education Center

- The award is given upon the completion of the project “Parallelization of bandit algorithms to reduce the computational cost of news article recommendation system.”

REFERENCES

Professor Seungryul Baek

Supervisor

Masters

- **Personal website:** <https://sites.google.com/site/bsrvision00/>
- **Affiliation:** Associate Professor, Artificial Intelligent Graduate School (AIGS), UNIST, South Korea
- **Affiliation:** Associate Professor, Department of Computer Science and Engineering, UNIST, South Korea
- **Email:** srbaek@unist.ac.kr

Professor Binod Bhattarai

Co-supervisor

Masters

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- **Affiliation:** Lecturer, University of Aberdeen, Aberdeen, UK
- **Affiliation:** Honorary Lecturer, University College London, UK
- **Email:** binod.bhattarai@abdn.ac.uk